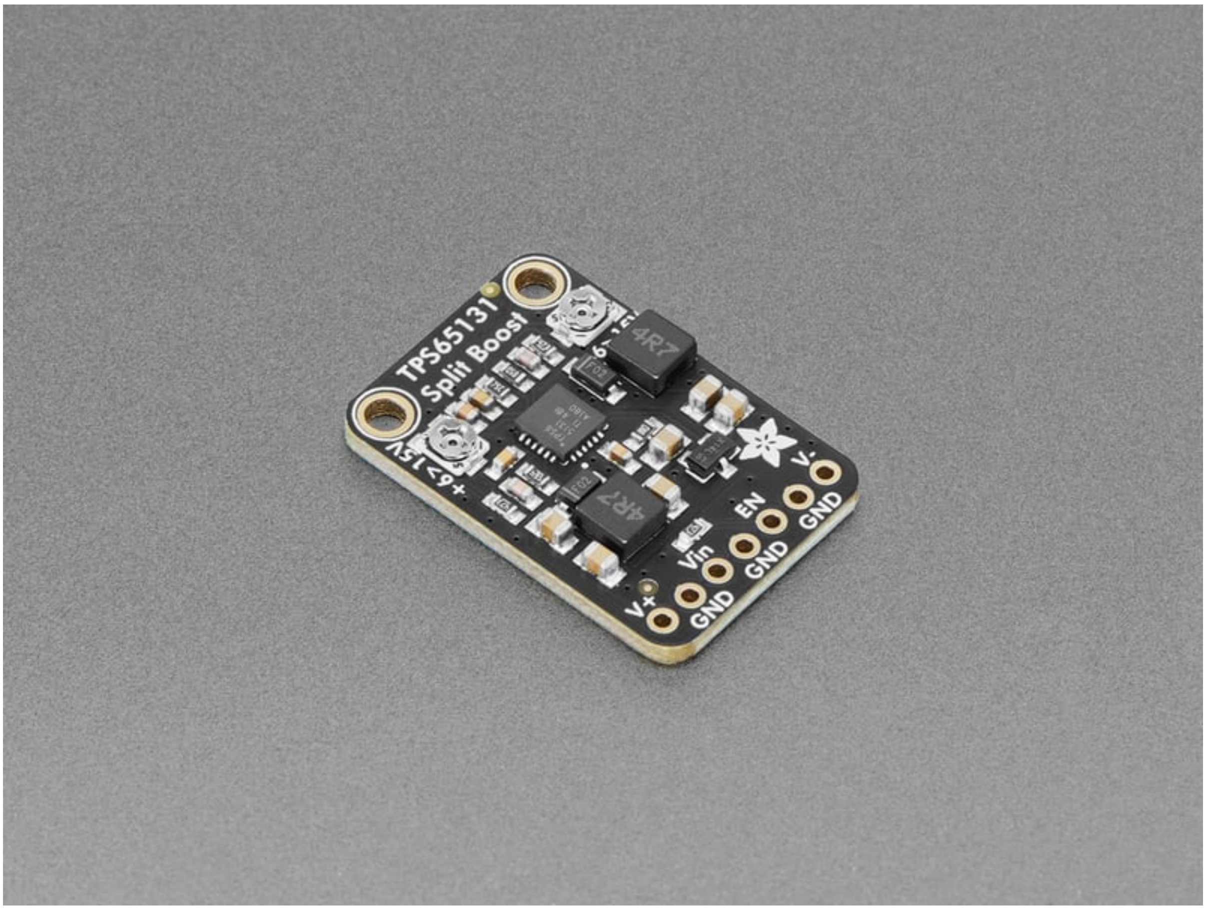
Adafruit TPS65131 Split Power Supply Boost Converter

Created by Liz Clark



https://learn.adafruit.com/adafruit-tps65131-split-power-supply-boost-converter

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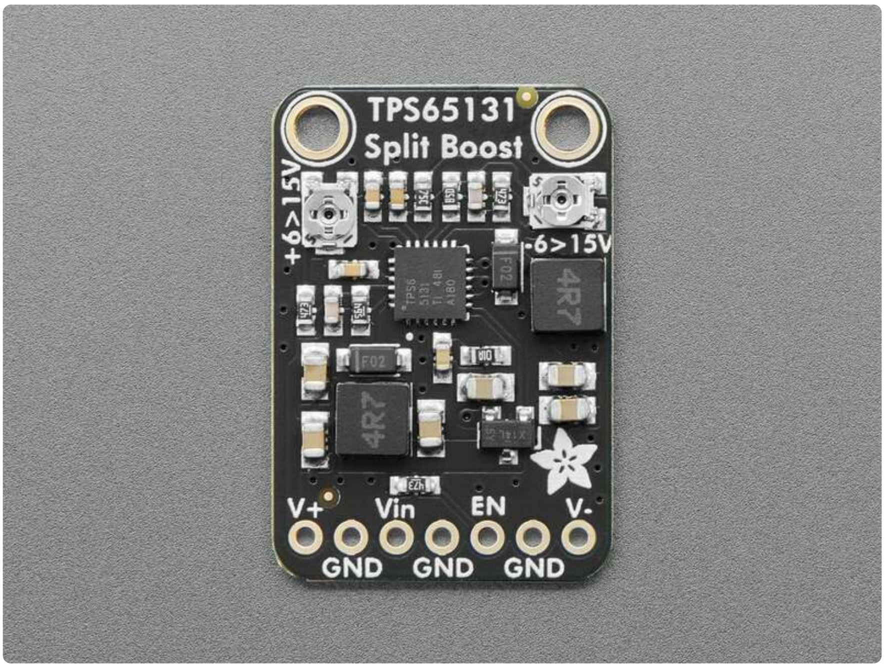
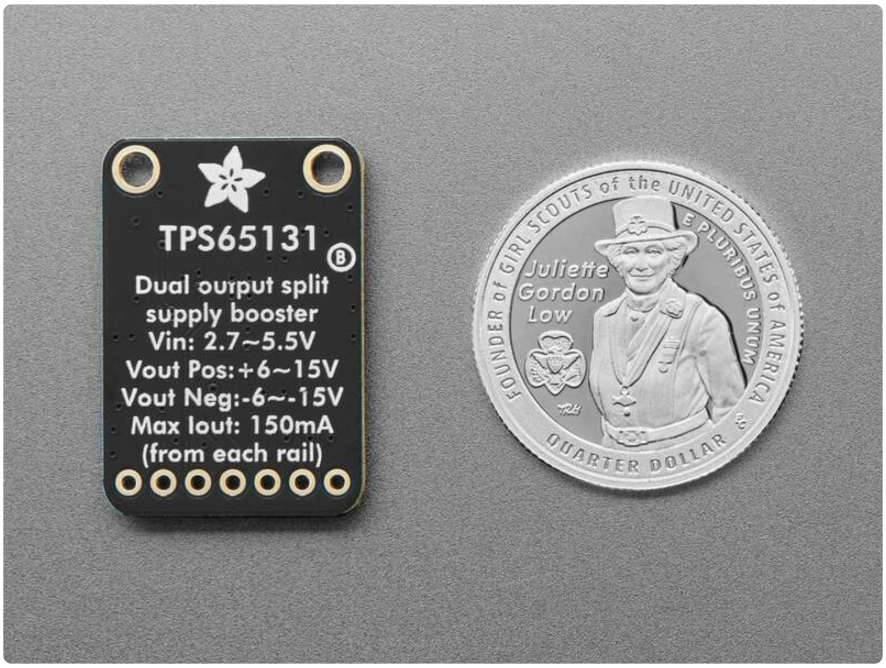
Overview

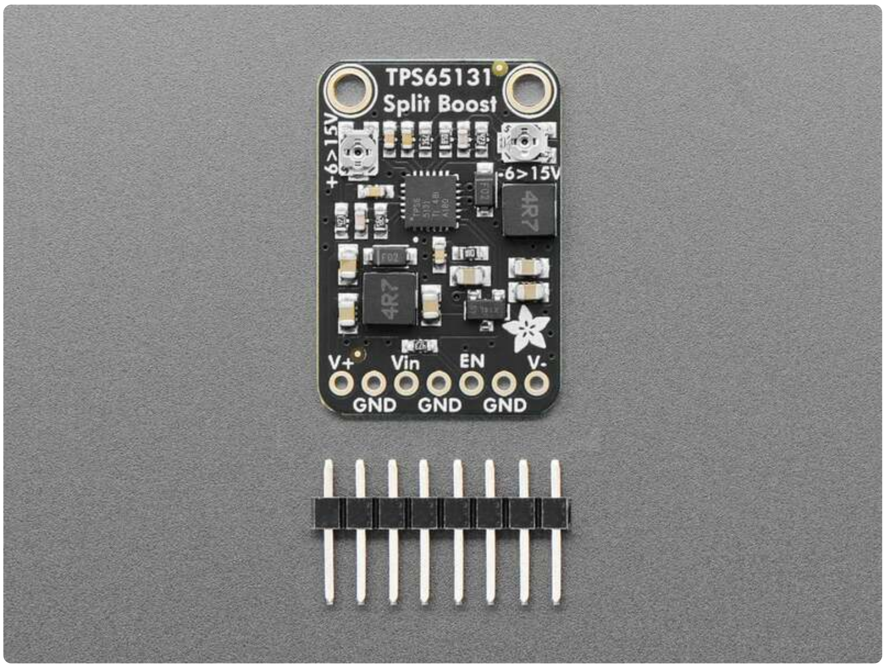


Need those tricky split-rail voltages for your vintage electronics? The **Adafruit TPS65131 Split Rail Boost Converter** generates both positive and negative voltage rails from a single 3-5V input. This makes it great for mimicking or replacing a transformer-based power supply or something that needs negative bias. We kept the design compact, and it even has adjustable output pots, so you can tune the voltages from about +6~+15VDC and -6~15VDC. Perfect for use with audio or retro projects now that transformer supplies are hard to get, or if you want to have something that could run on battery or USB.



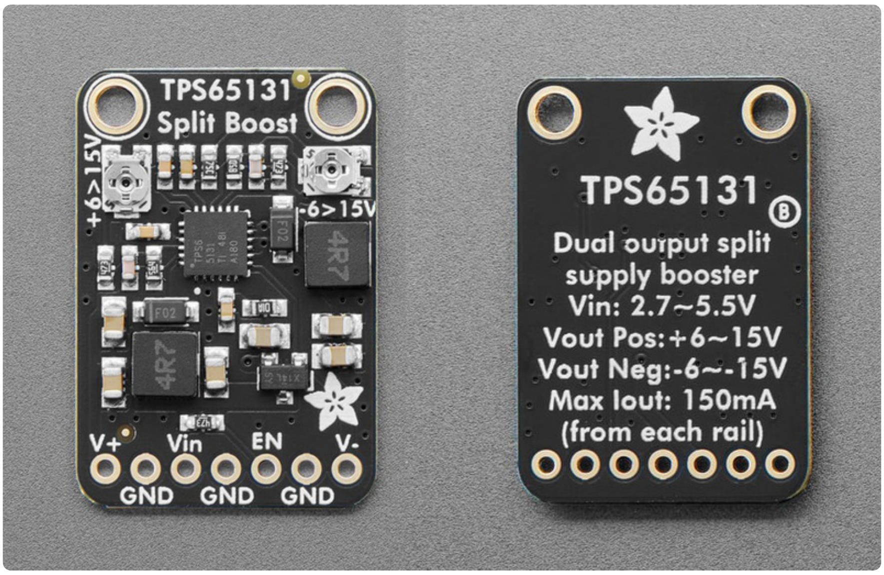
Features:

- The TPS65131 can generate positive and negative voltages simultaneously
- Adjustable voltage ranges: +6V to +15V and -6V to -15V, these are somewhat approximate because the trimmer pot can vary, you may find +/-0.5V variance on the bookend values
- Input voltage: 3 to 5.5V
- Output current: 150mA from each rail (assuming +12/-12V from 4.5V input). So not good for a motor, best for biasing or powering analog/digital electronics
- Compact breadboard-friendly design with mounting holes and two trimmer pots



We initially designed this to generate a -12V bias voltage on an Apple][disk drive, but you could definitely use it for synthesizers/modulars, TTL logic or op-amps that want a wide power supply, biasing LCDs or OLEDs, as well as other uncommon voltages. The trimmers let you tweak the outputs so you could get +6V and -12V or +12 and -8V.

Pinouts



- **Vin** - this is the power input pin. It can be powered with 3V to 5.5V.
- **V+** - this is the positive voltage output. It can range from +6V to +15V.
- **V-** - this is the negative voltage output. It can range from -6V to -15V.
- **GND** - common ground
- **EN** - this is the enable pin. You can tie this pin to **GND** to disable the TPS65131.

Trimmer Pots

Two trimmer pots let you dial in the voltage output on the positive (V+) and negative (V-) pins. Because the trimmer pots can vary, you may find +/-0.5V variance on the bookend values. Each output has a maximum output of 150mA.

- **Positive voltage trimmer pot** - On the left side of the board is the trimmer pot for the positive voltage output. It is labeled **+6>15V** on the board silk.
- **Negative voltage trimmer pot** - On the right side of the board is the trimmer pot for the negative voltage output. It is labeled **-6>15V** on the board silk.

Use

You can use a multimeter with the TPS65131 breakout to measure the output voltages. In the example below, you'll supply 5V to **VIN** and then use the multimeter to measure the positive voltage output and the negative voltage output. The outputs are adjusted with the onboard trim pots.

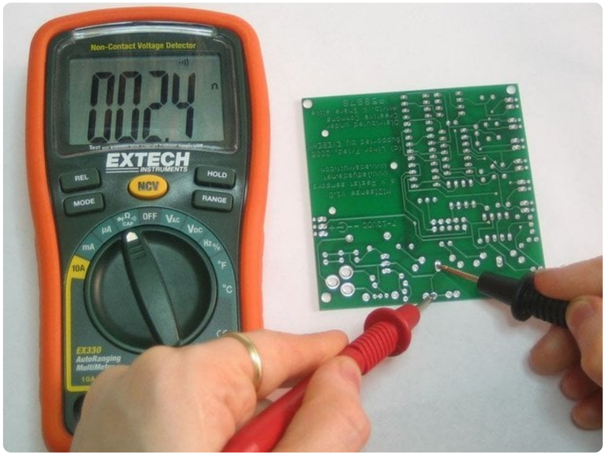
Pocket Autoranging Digital Multimeter
When we're on the go, we like to keep a multimeter in our purse and this model is by far the best pocket meter we've found. It's so good you'll end up using it as your...
https://www.adafruit.com/product/850

5V 2A (2000mA) switching power supply - UL Listed
This is an FCC/CE certified and UL listed power supply. Need a lot of 5V power? This switching supply gives a clean regulated 5V output at up to 2000mA. 110 or 240 input, so it works...
https://www.adafruit.com/product/276

Female DC Power adapter - 2.1mm jack to screw terminal block
If you need to connect a DC power wall wart to a board that doesn't have a DC jack - this adapter will come in very handy! There is a 2.1mm DC jack on one end, and a screw terminal...
https://www.adafruit.com/product/368

Multimeter Setup

Set your multimeter to the DC voltage mode. It is labeled with a V and two lines, one dashed and one solid. For information on using a multimeter, check out this guide. (https://adafru.it/192A)

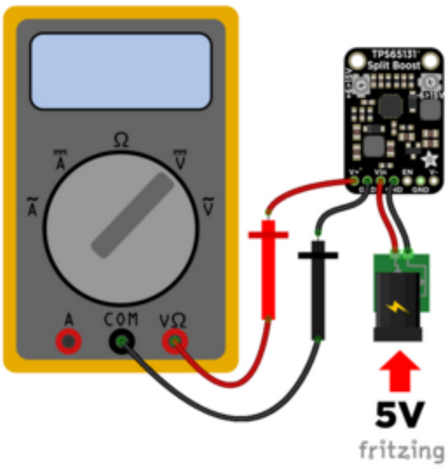


Multimeters
By lady ada
Overview

https://learn.adafruit.com/multimeters/overview

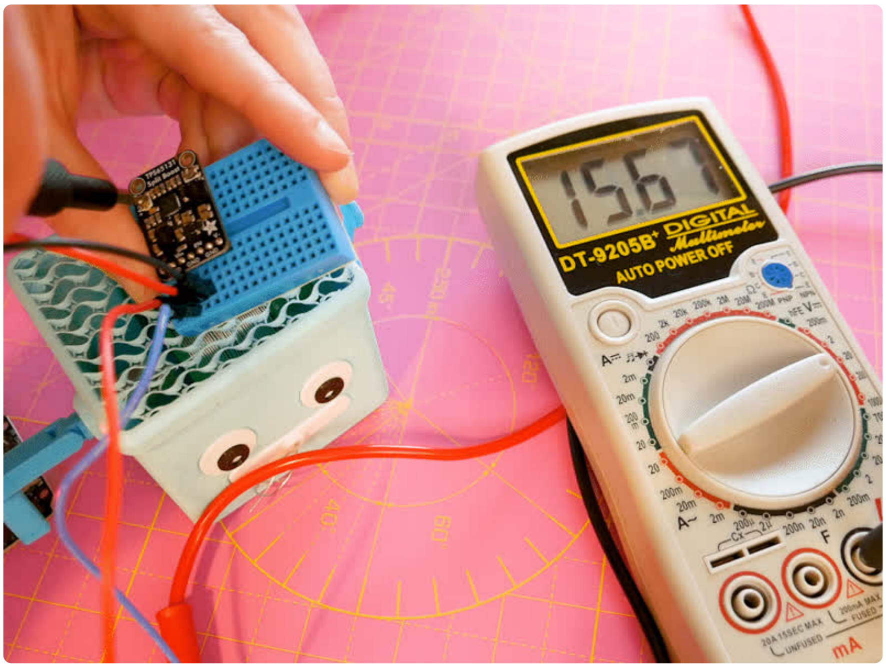
Measure Positive Voltage

Wire up the breakout to a 5V power supply and your multimeter as shown below:



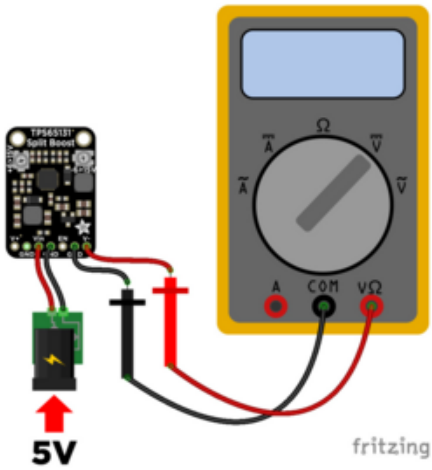
Breakout Vin to **5V power supply positive (red wire)**
Breakout GND to **power supply negative (black wire)**
Breakout V+ to **multimeter volts (red wire)**
Breakout GND to **multimeter COM (black wire)**

After wiring and powering everything up, adjust the voltage output on the V+ pin with the positive voltage trim pot on the left side of the board. As you turn the pot clockwise, you'll see the voltage increase. As you turn the pot counter-clockwise, you'll see the voltage decrease. You can tune the voltages from about +6~+15VDC.



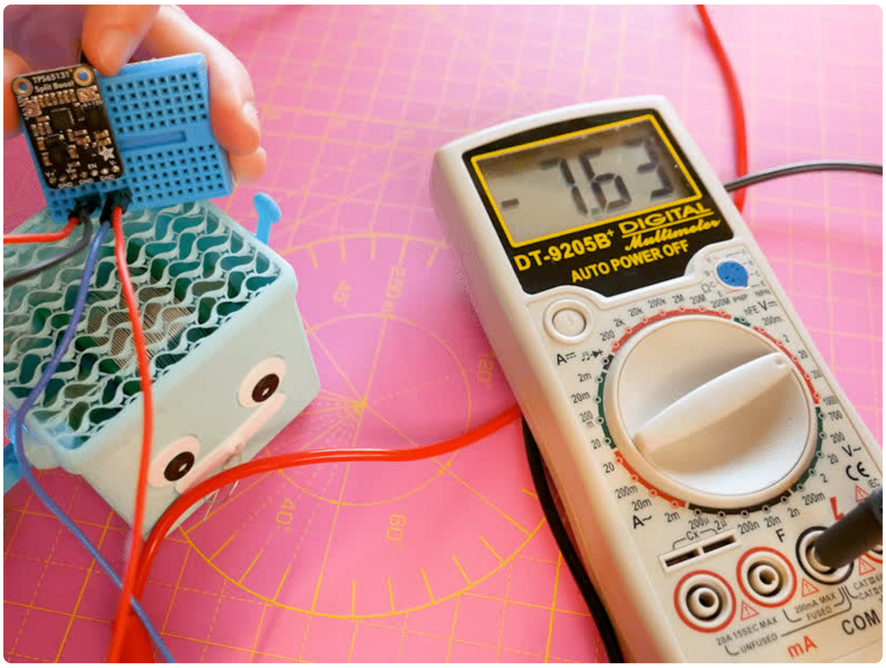
Measure Negative Voltage

Wire up the breakout to a 5V power supply and your multimeter as shown below:



Breakout Vin to **5V power supply positive (red wire)**
Breakout GND to **power supply negative (black wire)**
Breakout V- to **multimeter volts (red wire)**
Breakout GND to **multimeter COM (black wire)**

After wiring and powering everything up, adjust the voltage output on the V- pin with the negative voltage trim pot on the right side of the board. As you turn the pot clockwise, you'll see the negative voltage increase. As you turn the pot counter-clockwise, you'll see the negative voltage decrease. You can tune the voltages from about -6~15VDC.

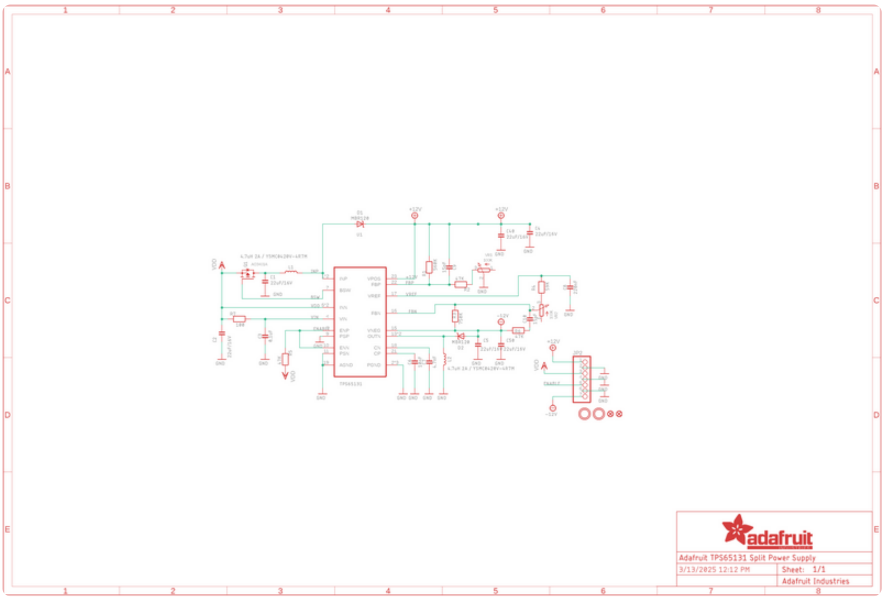


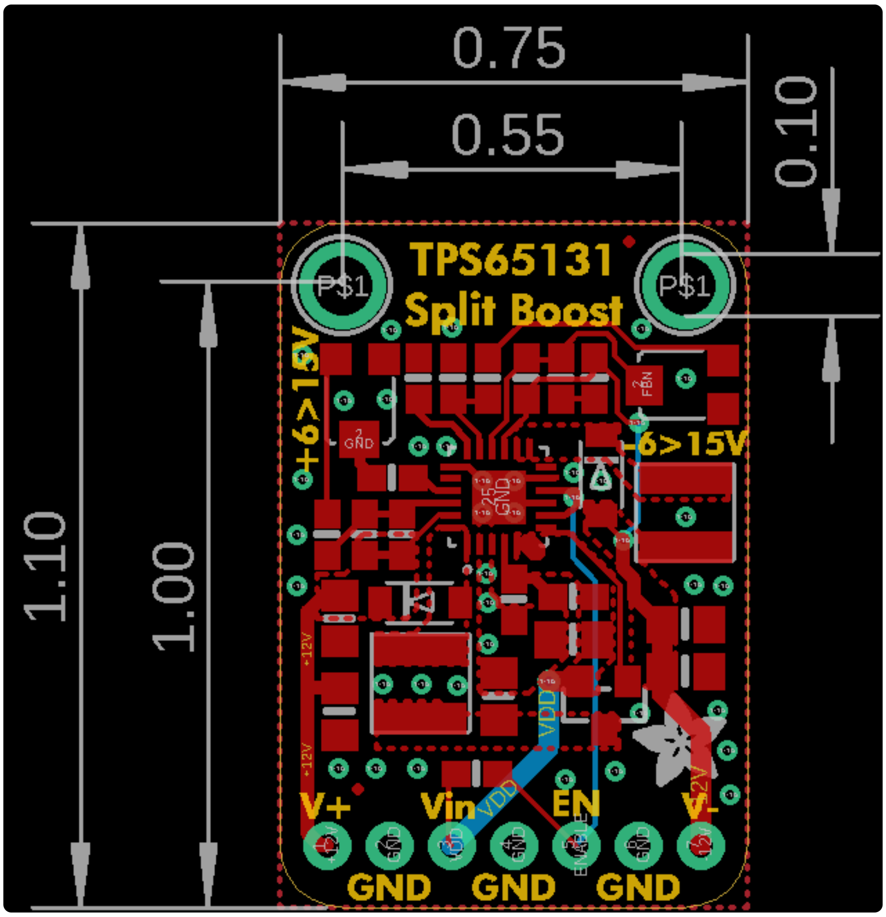
Downloads

Files

- [TPS65131 Datasheet](https://adafru.it/1afS) (https://adafru.it/1afS)
- [EagleCAD PCB Files on GitHub](https://adafru.it/1afT) (https://adafru.it/1afT)
- [Fritzing object in the Adafruit Fritzing Library](https://adafru.it/1afU) (https://adafru.it/1afU)

Schematic and Fab Print




0.75
0.55
0.10
1.10
1.00
TPS65131
Split Boost
+6>15V
-6>15V
V+
Vin
EN
V-
GND
GND
GND